



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
25-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-2

Time : 3 Hours

Maximum Marks : 204

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **20** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

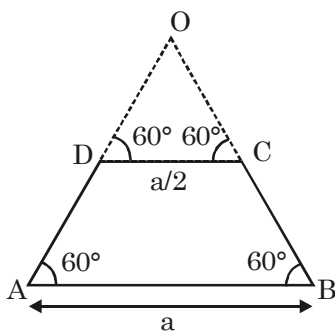
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

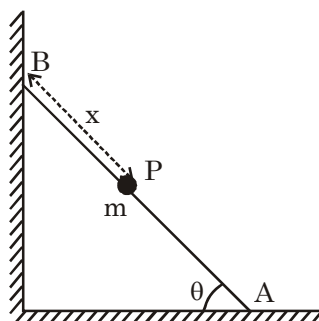
1. Consider a uniformly charged sheet ABCD, which is a part of an equilateral triangular sheet of side a as shown in the figure. Choose the **CORRECT** options regarding the electric field E at point O due to this sheet.



- (A) Magnitude of $\vec{E}$, increases with the increase in a (Keeping charge density same)
 (B) Magnitude of E , decreases with increase in a (keeping total charge same)
 (C) If charge density is σ and $a = 1$ m, magnitude of E is equal to $\frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$.
 (D) If charge density is σ and $a = 2$ m, magnitude of E is equal to $\frac{7\sigma}{22\epsilon_0} \ln \sqrt{2}$.

Space for Rough Work

2. A light stick of length ℓ rests with its one end against the smooth wall and other end against the smooth horizontal floor as shown in the figure. The bug starts at rest from point B and moves such that the stick always remains at rest, a_p is the magnitude of acceleration of bug of mass m , which depends upon its distance of x from the top end of the stick. Choose the **CORRECT** option(s) :



(A) $a_p = \frac{g}{\sin \theta} \left(1 - \frac{x}{\ell} \right)$

(B) $a_p = \frac{g}{\cos \theta} \left(1 - \frac{x}{\ell} \right)$

- (C) The time taken by the bug to reach the bottom of the stick having started at the top end

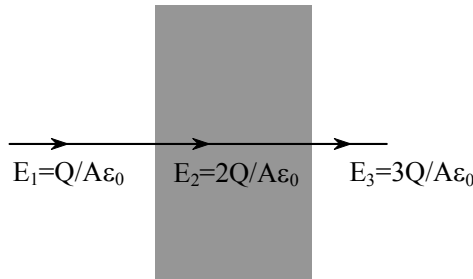
from rest is $\frac{\pi}{2} \sqrt{\frac{\ell \sin \theta}{3g}}$.

- (D) The time taken by the bug to reach the bottom of the stick having started at the top end

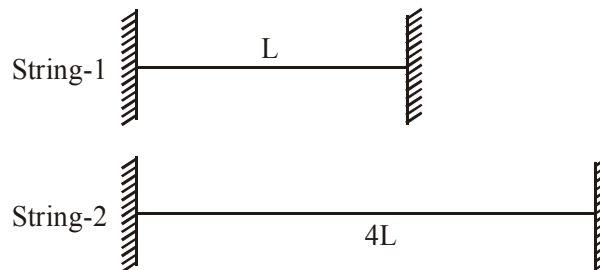
from rest is $\frac{\pi}{2} \sqrt{\frac{\ell \sin \theta}{g}}$.

Space for Rough Work

3. A very large slab of insulating material having face area A and dielectric constant $K = 2$ is placed between plates of parallel plate capacitor and some uniform charges on plates are given. The system is placed in uniform electric field (fig). The field is perpendicular to the surface of the material.



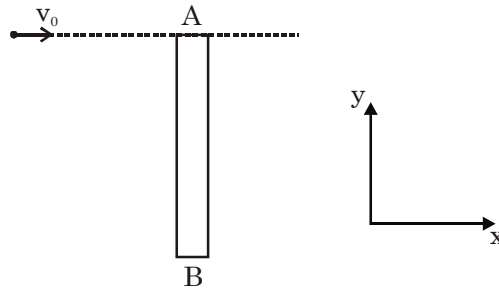
- (A) The total charge on right surface is Q
 (B) The free charge on left surface is $3Q$
 (C) The bound charge on right surface is $2Q$
 (D) The magnitude of the applied electric field is $2Q/A\epsilon_0$.
4. Two strings made of same material, same cross-section but different lengths are fixed at both ends and excited in their respective fundamental modes such that total oscillation energy is the same in both cases. Let f_1, f_2 be the oscillation frequencies and A_1, A_2 be the amplitudes of oscillation of the antinodes in the two cases. Assuming both strings to be under same tension, pick the **CORRECT** option(s).



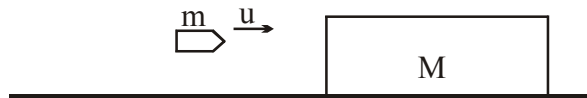
- (A) $\frac{f_1}{f_2} = \frac{1}{4}$ (B) $\frac{A_1}{A_2} = \frac{1}{2}$ (C) $\frac{f_1}{f_2} = \frac{4}{1}$ (D) $\frac{A_1}{A_2} = \frac{2}{1}$

Space for Rough Work

5. A uniform rod of mass m and length ℓ lies on a horizontal frictionless table. A particle of same mass moving with a speed v_0 sticks to one end of the rod at $t = 0$. At $t = t$



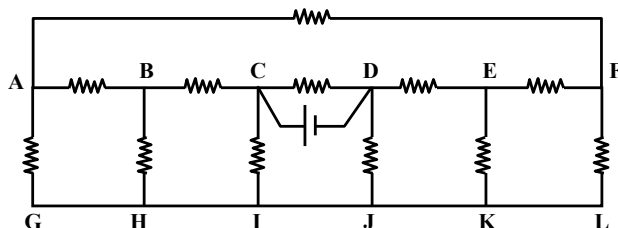
- (A) force exerted by particle on rod will be $\frac{9}{25} \frac{mv_0^2}{\ell}$
- (B) just after collision acceleration of point B of rod in y-direction is $\frac{25}{27} \frac{v_0^2}{\ell}$
- (C) In the frame of centre of mass of system, path of centre of the rod is circle.
- (D) Path of centre of rod is cycloid.
6. A bullet of mass m travelling horizontally with speed u strikes a wooden block of mass ' M ' placed on a smooth horizontal plane. The penetration is assumed uniform and the bullet comes to rest with respect to block after penetrating a distance ' d ' into the block. Then choose the **CORRECT** statement(s):



- (A) Ultimately the total loss of kinetic energy is $\frac{mMu^2}{2(M+m)}$
- (B) The value of the resistance force (assumed constant) offered by the wood is $F = \frac{mMu^2}{2d(M+M)}$
- (C) The distance covered by the bullet w.r.t. the ground before it comes to rest w.r.t. block is not equal to $\frac{dm}{M+m}$
- (D) The block moves less distance than the bullet wrt ground.

Space for Rough Work

7. Hydrogen gas absorbs radiations of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which three wavelengths are shorter than λ_0 . Choose the **CORRECT** statement(s).
- (A) The final excited state of the atom is $n = 4$
 (B) The initial state of the atom may be $n = 2$
 (C) There are two transitions in balmer series
 (D) There are three transitions belonging to Lyman series.
8. In the figure shown, resistance of each resistor are same. If current in AG is 5A then



- (A) current in FL is 5 A
 (B) current in BA is 15 A
 (C) Current in AF from top resistor is 10 A
 (D) current in BH is 20 A

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

9. A particle is taken to a distance r ($> R$) from centre of earth. R is the radius of earth. It is given velocity v which is perpendicular to $\vec{r}$ (measured from centre of earth). Match the following assuming particle do not collide with earth. Mass of particle (m) is negligible in comparison to mass of earth (M) and assume there is no air resistance :

List-I

List-II

- | | |
|---|--|
| (P) $v = \sqrt{\frac{GM}{r}}$ | (1) Distance of point mass from centre of earth first increases then decreases |
| (Q) $v = \sqrt{\frac{2GM}{r}}$ | (2) Distance of point mass from centre of earth first decreases then increases |
| (R) $\sqrt{\frac{2GM}{r}} > v > \sqrt{\frac{GM}{r}}$ | (3) Distance of point mass from centre of earth monotonically increases |
| (S) $\sqrt{\frac{GM}{r}} > v > \sqrt{\frac{GM}{r} \left(\frac{2R}{r+R} \right)}$ | (4) Distance of point mass from centre of earth remain constant |

(A) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 2$

(B) $P \rightarrow 3$; $Q \rightarrow 4$; $R \rightarrow 2$; $S \rightarrow 1$

(C) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 4$

(D) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 1$

Space for Rough Work

10. Three ants are sitting on the vertices of an equilateral triangle of side of a . At $t = 0$, ant 1 starts approaching ant 2 with a speed v , ant 2 starts approaching ant 3 with a speed v and ant 3 start approaching ant 1 with a speed v . If $a = 1\text{m}$, $v = 1\text{m/s}$ match the following

List-I

List-II

- | | |
|---|------------------------------|
| (P) The three ant meet at the centroid of the triangle at time (in s) | (1) $\frac{\sqrt{3}}{2}$ |
| (Q) The angular velocity (in rad/s) of ant 2 with respect to ant 1 at $t = 0$ is | (2) $\frac{25\sqrt{3}}{256}$ |
| (R) The angular acceleration (in rad/s^2) of ant 2 with respect to ant 1 at $t = 0$ is | (3) $\frac{2}{3}$ |
| (S) The area of the equilateral triangle (in m^2) formed by the three ants at any time $t = a/4v$ will be | (4) $\frac{3\sqrt{3}}{4}$ |

(A) $P \rightarrow 1 ; Q \rightarrow 4 ; R \rightarrow 2 ; S \rightarrow 3$

(B) $P \rightarrow 3 ; Q \rightarrow 1 ; R \rightarrow 4 ; S \rightarrow 2$

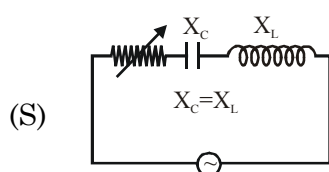
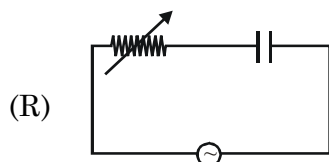
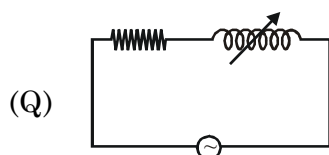
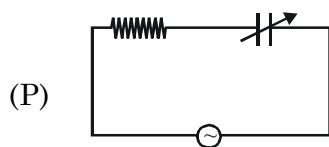
(C) $P \rightarrow 4 ; Q \rightarrow 3 ; R \rightarrow 1 ; S \rightarrow 2$

(D) $P \rightarrow 3 ; Q \rightarrow 1 ; R \rightarrow 1 ; S \rightarrow 4$

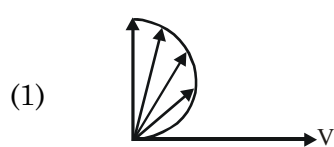
Space for Rough Work

11. In a series AC circuit if we change either resistance or reactance not only current amplitude changes, but phase angle between current and voltage also changes. If we take voltage as reference and draw current phasor then head of this phasor moves on a particular trajectory. In List-I, variation of resistance or reactance is specified and in List-II is given path followed by head of current phasor.

List-I



List-II



(A) $P \rightarrow 1$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 4$

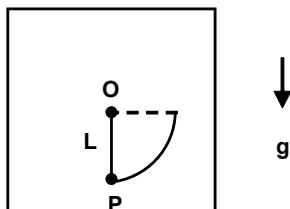
(B) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 2$

(C) $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 4$

(D) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$

Space for Rough Work

12. Figure shows a box B of mass m held fixed at large height H (which is very large in comparison to dimension of the box, but very small in comparison to radius of earth). A particle P whose mass is equal to that of box is suspended at O (which is centre of mass of the box) with a massless string. Particle can swing in vertical plane without colliding box. Particle is pulled aside to horizontal position and released. The box is held fixed till the particle reaches lowest position and then box is released at $t = 0$. t is a general time before box collides ground.



List-I

List-II

- (P) Tension in the thread at the $t = t$ in the units of mg will be (1) 1
- (Q) Speed of box at time $t = \pi\sqrt{\frac{\ell}{2g}}$ in the units of $\sqrt{\frac{g\ell}{8}}\sqrt{\pi^2 + 4}$ will be (2) 2
- (R) Acceleration of the box at $t = \pi\sqrt{\frac{\ell}{2g}}$ in the units of g will be (3) 0
- (S) Initial acceleration ($t = 0^+$) of particle in the units of g will be (4) none
- (A) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$ (B) $P \rightarrow 1$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 4$
- (C) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 3$; $S \rightarrow 2$ (D) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

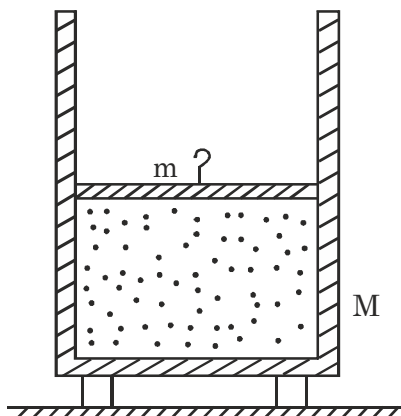
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	○	○	○	○
①	①	①	①	①	①
②	②	②	●	②	②
③	③	③	③	③	③
④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	●
⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨

		-			
●	●	○	○	○	○
①	①	①	①	①	①
②	②	②	②	●	②
③	③	③	③	③	③
④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨

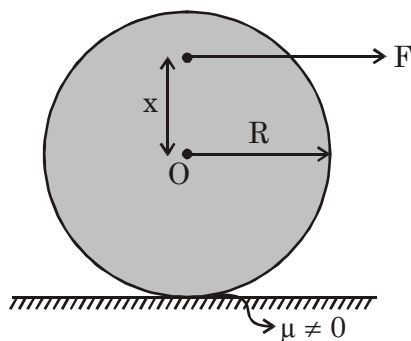
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- As shown, a piston chamber of cross section area A is filled with ideal gas. A sealed piston of mass m is right at the middle height of the cylinder at equilibrium. The friction force between the chamber wall and the piston can be ignored. The mass of the rest of the chamber is M . The atmosphere pressure is P_0 . Now slowly pull the piston upwards isothermally, find the maximum value of $\frac{M}{m}$ such that the chamber can be lifted off the ground. Given that $P_0 = \frac{3mg}{2A}$



Space for Rough Work

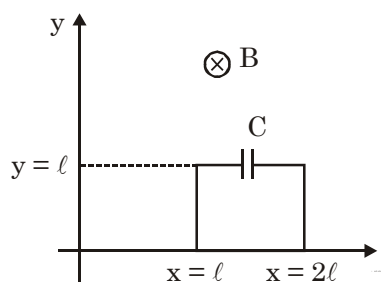
2. Initially a disc of radius 30 cm was in rest & we apply force F as shown in figure. If friction acting on disc is zero then find the value of x (in cm).



3. A glass bulb with volumetric expansion coefficient $\alpha = 1.50 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ is weighed in liquid at temperatures $t = 10^\circ\text{C}$ and $t_1 = 20^\circ\text{C}$. The weights of the displaced liquids are $p = 100.01 \text{ g}$ and $p_1 = 100 \text{ g}$ respectively. Find the volumetric expansion coefficient α_1 (in the units of $10^{-5} \text{ } ^\circ\text{C}^{-1}$) of the liquid in the temperature interval from t to t_1 .

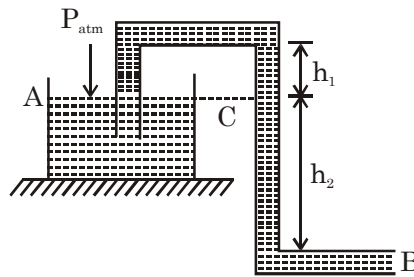
Space for Rough Work

4. A mercury drop of radius 1 cm is sprayed into 10^6 drops of equal size. Calculate the energy (in mJ) spent for the process (Surface Tension of Mercury = $35 \times 10^{-3} \text{ Nm}^{-1}$).
5. One end of a string of length $L = 15\text{m}$ is tied to the ceiling of an elevator accelerating upwards with an acceleration 20 m/s^2 . The other end of the string is free. The linear mass density of the string varies linearly from 0 to λ_0 from bottom to top. Find the time taken by a pulse (in seconds) to reach from bottom to the top. ($g = 10 \text{ ms}^{-2}$)
6. A square frame of wire connected to a capacitor is kept in a magnetic field which varies with position as well as time and is given as $\vec{B} = \alpha x t (-\hat{k})$ as shown in figure, where α is a constant. The charge on the capacitor at any time is $\frac{n}{2} C \alpha \ell^3$, Find n.



Space for Rough Work

7. A radioactive sample has decay constant λ . The rate of production of nuclei in the given sample is $\frac{9\lambda N_0^2}{N}$, where N_0 is the number of radioactive nuclei in the sample at $t = 0$ and N is the number of radioactive nuclei in the sample at time $t = t$ sec. If the number of nuclei present in the radioactive sample at $t \rightarrow \infty$ is $m \times 10^6$ nuclei then find the value of m . (Given $N_0 = 10^6$ nuclei)
8. A tube of uniform cross-section is used to siphon water from a vessel, as shown in the Figure. Assuming that the cross-section of the vessel is very much greater than that of the tube, find the maximum value of height h_1 (in m), for which the siphon will work. [Take $h_2 = 3$ m, atmospheric pressure $P_{\text{atm}} = 10^5 \text{ Nm}^{-2}$, density of water $\rho = 1000 \text{ kg}\cdot\text{m}^{-3}$ and acceleration due to gravity $g = 10 \text{ ms}^{-2}$]



Space for Rough Work

PART-2 : CHEMISTRY

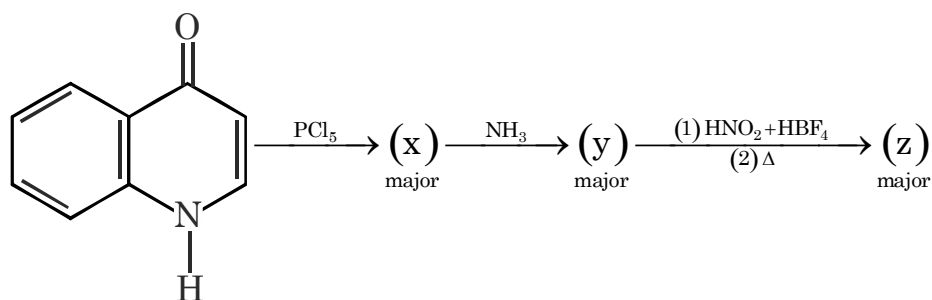
SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

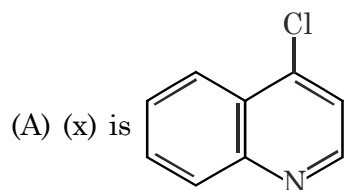
1. Which of the following produce(s) gaseous product on treatment with dilute HCl ?
(A) $\text{Na}_2\text{S}_2\text{O}_3$ (B) NaHCO_3 (C) Na_2SO_3 (D) Na_2S
2. The complex(s) which can be oxidised on exposure into air or on boiling is / are
(A) $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ (B) $\text{K}_4[\text{Co}(\text{CN})_6]$ (C) $\text{Na}_4[\text{Co}(\text{NO}_2)_6]$ (D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

Space for Rough Work

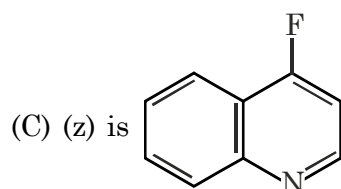
3. Consider the reaction given below :



Select the correct statement(s)

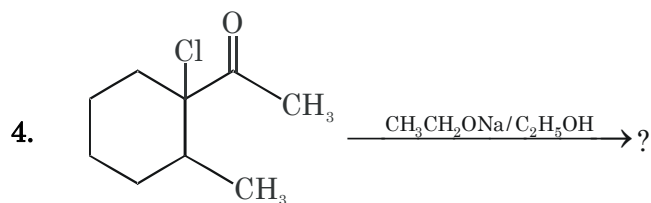


(B) The product (y) is gives carbayl amine test



(D) conversion of (x) into (y) involves nucleophilic substitution

Space for Rough Work

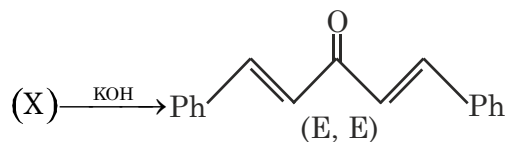


Select the correct statement(s)

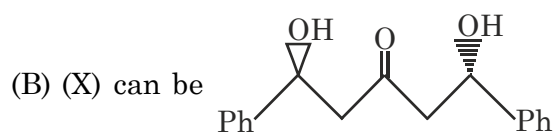
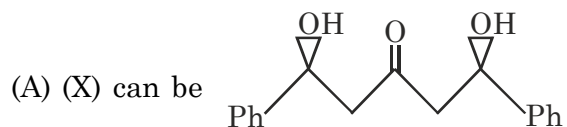
- (A) The major product formed is an ester
 (B) The given reaction involves favorskii rearrangement
 (C) Carbocationic intermediate is involved
 (D) The major product is an ether
5. An aqueous solution of 6.3gm of a hydrated oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot x\text{H}_2\text{O}$) is made up to 250ml. The 40ml of 0.1N NaOH was required to completely neutralize 10ml of the above prepared stock solution. Which of the following statement(s) are correct ?
- (A) The acid is dihydrated
 (B) Equivalent weight of the hydrated salt is 45
 (C) Equivalent weight of anhydrous acid is 45
 (D) 20 ml of the same stock solution would require 40ml 0.1M $\text{Ca}(\text{OH})_2$ for complete neutralisation
6. Which of the following statement is correct ?
- (A) For any cyclic process, $\Delta U = 0$
 (B) $\left(\frac{\delta U}{\delta V}\right)_T = \frac{a}{V^2}$ for 1 mole of a gas obeying van der Waal's equation
 (C) $\bar{C}_p = \infty$ for water when it is in equilibrium with vapour at 1 atm pressure and 373 K.
 (D) For any real gas, the molar internal energy can be given as $\frac{3}{2}R$

Space for Rough Work

7. Which of the following reagent(s) would be most suitable to convert $(\text{CH}_3)_3\text{N}$ into $(\text{CH}_3)_2\text{NH}$
- (A) HCl only (B) NCBr
- (C) $\text{H}_3\text{O}^+; \Delta$ (D) $(\text{PPh}_3 + \text{H}_2\text{O})$ only
8. Consider the reaction given below :-



Select the correct statement(s)



- (C) The reaction involves E1_{CB}
- (D) The reaction involves E2 mechanism

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

9. List – I

(Metals)

(P) Fe

(Q) Cu

(R) Mg

(S) Au

List – II

(Method of extration)

(1) Self reduction

(2) Carbon reduction

(3) Cyanide process

(4) Electrolytic reduction

The correct option is -

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$ (B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$ (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$ (D) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

Space for Rough Work

10. List – I

 (P) PF_5

 (Q) SF_6

 (R) IF_7

 (S) XeF_8^{2-}
List – II

(1) Equatorial bond longer than axial bond

(2) Axial bond longer than equatorial bond

(3) All bonds are of equal length

(4) s-orbital is stereochemically inactive

The correct option is -

 (A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 1; \text{S} \rightarrow 4$

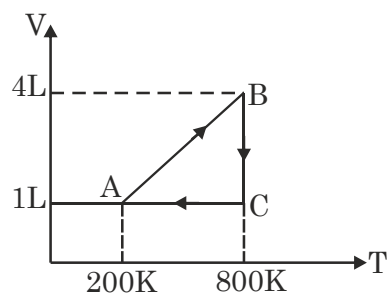
 (B) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

 (C) $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 2$

 (D) $\text{P} \rightarrow 3; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 4$

Space for Rough Work

11. Two mole of an ideal gas undergoes a cyclic ABCA as shown in the diagram below. However, it is known that during the process AB, the gas uses only 25% of energy absorbed as heat, in doing work.



List – I

- (P) Magnitude of ΔU for process AB
 (Q) Magnitude of q for process BC
 (R) Magnitude of ΔH for process CA
 (S) Magnitude of total work done during cycle

List – II

- (1) 9.6 KCal
 (2) 4.43 KCal
 (3) 2.03 KCal
 (4) 7.2 Kcal

The correct option is -

- (A) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 3; S $\rightarrow$ 4 (B) P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 1; S $\rightarrow$ 3
 (C) P $\rightarrow$ 4; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 3 (D) P $\rightarrow$ 4; Q $\rightarrow$ 1; R $\rightarrow$ 2; S $\rightarrow$ 3

Space for Rough Work

12. Given the following data

$$\Delta_f H^\circ (\text{CH}_4, \text{g}) = -80 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\circ (\text{C}_2\text{H}_6, \text{g}) = -85 \text{ kJ mol}^{-1}$$

$$\Delta_f H^\circ (\text{C}_2\text{H}_4, \text{g}) = 50 \text{ kJ mol}^{-1}$$

$$\text{Enthalpy of sublimation of Carbon (graphite)} = 720 \text{ kJ mol}^{-1}$$

$$\text{Enthalpy of atomisation of H} = 215 \text{ kJ mol}^{-1}$$

List – I

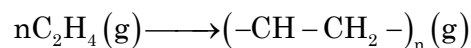
(Thermodynamic data)

(P) Bond enthalpy of C–H bond (per mole)

(Q) Bond enthalpy of C–C bond (per mole)

(R) Bond enthalpy of C=C bond (per mole)

(S) ΔH° , for the reaction :



(for 10 moles of ethene)

The correct option is -

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

List – II

(Values with respect to magnitude)

(1) 325 kJ

(2) 600 kJ

(3) 415 kJ

(4) 590 kJ

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

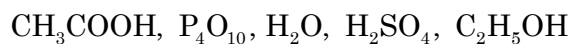
-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

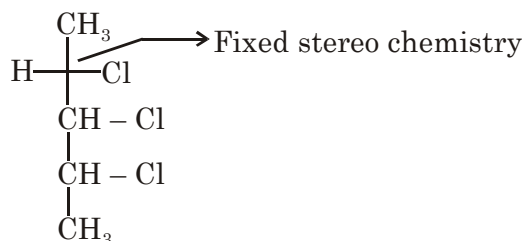
1. Equivalent conductance of 0.2M aqueous solution of a weak monobasic acid HA is $10\text{cm}^2\text{eq}^{-1}$ and that at infinite dilution is $200\text{cm}^2\text{eq}^{-1}$. What is the concentration of H^+ ions.
2. A definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a fixed temperature and 0.1atm pressure. After the equilibrium is established, total pressure inside the flask is 1.1atm. Value of K_p in $(\text{atm})^2$ is :
3. Density of $\text{X}_{(\text{g})}$ at 1 atm pressure and 500K is 0.8Kg/m^3 . It effuses through a small hole at a rate of $4/5$ times that of O_2 and at same condition find the compressibility factor of vapour.
4. Consider the given complex ion $[\text{Pt}(\text{bn})_2]^{2+}$, where bn stands for butylene diamine. Find the value of $\frac{n}{2}$ where n is the total number of stereoisomers.

Space for Rough Work

5. Find the value of $\frac{n}{4}$, where n is the number of compounds will produce POCl_3 when treated with PCl_5 as intermediate product or final product.



6. Calculate total number of optically active stereoisomers for following compound?



7. Calculate the specific rotation of (+) – glyceraldehyde, if a solution containing 2.0 gm/ 10ml is placed in a 1 gm polarimeter tube and its observed rotation at 25°C using sodium D line is +1.74°.
8. Calculate the isoelectric point (pI) of lysine. Give the pK_a of $\alpha\text{-NH}_3$ is 8.95, pK_a side chain NH_3 is 10.53 and pK_a of $\alpha\text{-COOH}$ is 2.18.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let O be the circumcentre of acute angled $\triangle ABC$ and let r_1, r_2 and r_3 be the radii of the circles drawn on the altitudes OD, OE and OF of triangle OBC, OCA and OAB as diameter,

then minimum value of $\frac{a^2}{r_1^2} + \frac{b^2}{r_2^2} + \frac{c^2}{r_3^2}$ is divisible by :

- (A) 36 (B) 72 (C) 144 (D) 216

2. $f : [0, 1] \rightarrow R$ is a function such that $f'(x)$ is continuous and $\int_0^1 f(x) dx = 0$, if maximum value

of $|f'(x)|$ when $x \in [0, 1]$ is 24, the possible value of $\left| \int_0^x f(t) dt \right|$ $x \in [0, 1]$ is

- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

3. Let A and B are non singular matrices of order three, each of them commute with matrix X, where $AX = B$ and $XB = A$. If X is not an identity matrix, then $\det(X^4 + X^2 + I)$ is divisible by (where I is identity matrix of order three)
- (A) 36 (B) 27 (C) 9 (D) 3
4. Suppose that two complex number $z = a + ib$; $w = c + id$ satisfy the equation $\frac{z+w}{z} = \frac{w}{z+w}$, then (where a, b, c, d $\in \mathbb{R}$ and $\sqrt{-1} = i$)
- (A) $\arg z - \arg w = \frac{2\pi}{3}$ (B) both b and d are zero
- (C) $\arg z - \arg w = -\frac{2\pi}{3}$ (D) at least one of b and d is non-zero
5. Let $a_n = \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$. Then
- (A) a_n is less than $\int_1^n \frac{dx}{x}$ (B) a_n is greater than $\int_2^n \frac{dx}{x}$
- (C) $\lim_{n \rightarrow \infty} \frac{a_n}{\ln n} = 1$ (D) $\lim_{n \rightarrow \infty} a_n$ is finite

Space for Rough Work

6. Let $\alpha(x)$ and $\beta(x)$ be differentiable non linear functions on $[0, \infty)$ such that $\alpha(0) = 1$, $\beta(0) = 3$, $\beta(x) > 2\alpha(x)$, $2\alpha(x) + \beta'(x) = 5$ and $2\alpha'(x) + \beta(x) = 5$ for all $x \in [0, \infty)$, then
 (A) $\alpha(x)$ has one maxima in $[0, \infty)$ (B) $\alpha(x)$ has one minima in $[1, \infty)$
 (C) $\beta(x)$ is increasing function in $[0, \infty)$ (D) $\beta(x)$ is increasing function in $[1, \infty)$
7. Let G be the centroid of the ΔABC whose sides are of length a, b, c . If P be a point in the plane of ΔABC such that $PA = 1, PB = 3, PC = 4$ and $PG = 2$, then the value of $a^2 + b^2 + c^2$ is less than :
 (A) 40 (B) 44 (C) 46 (D) 48
8. If y_n is n th derivative of y w.r.t. x then
 (A) $y = (ax + b)^m$ then $y_n = {}^mP_n a^n (ax + b)^{m-n}$
 (B) $y = \log(ax + b)$ then $y_n = (-1)^{n-1} \frac{(n-1)! a^n}{(ax + b)^n}$
 (C) $y = \sin(ax + b)$ then $y_n = a^n \sin\left(ax + b + n\frac{\pi}{2}\right)$
 (D) $y = \cos(ax + b)$ then $y_n = a^n \cos\left(ax + bx - n\frac{\pi}{2}\right)$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

9. Match List – I with List – II and select the correct answer using the code given below the list.

List – I

(P) $\sum_{r=1}^n \frac{2r+1}{(r^2+r)^2}$

(Q) $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{(2r+1)}{(r^2+r)^2}$

(R) $\prod_{k=1}^n \left(\sum_{r=1}^k \frac{2r+1}{(r^2+r)^2} \right)$

(S) $\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(\sum_{r=1}^k \frac{2r+1}{(r^2+r)^2} \right)$

List – II

(1) $\frac{n+2}{2(n+1)}$

(2) $\frac{n(n+2)}{(n+1)^2}$

(3) 1

(4) $\frac{1}{2}$

(A) P → 2; Q → 3; R → 1; S → 4

(B) P → 3; Q → 1; R → 2; S → 4

(C) P → 1; Q → 3; R → 2; S → 4

(D) P → 4; Q → 1; R → 2; S → 3

Space for Rough Work

10. Match List-I with List-II and select the correct answer using the code given below the list.
If $y + x = 4$ is a tangent to the hyperbola at the point $(2, 2)$ intersects asymptotes of hyperbola at A and B. If length of AB is $4\sqrt{2}$ and centre of hyperbola is at origin, then match the following List-I with List-II

List-I

List-II

- (P) The equation of conjugate axis of hyperbola is
(Q) The equation of hyperbola is
(R) The equation of directrix can be
(S) The equation of transverse axis is
(A) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 2$
(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$
(D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
11. Consider four distinct circles with radii 3, r_1, r_2, r_3 ($r_1, r_2, r_3 > 3$) touches all three fixed distinct lines in a plane, where r_1, r_2, r_3 are roots of equation $x^3 - mx^2 + 243x - n = 0$, $m, n \in \mathbb{R}$

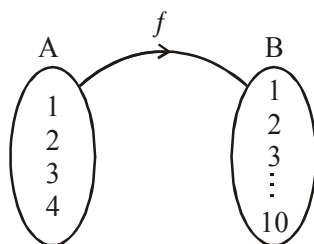
List-I

List-II

- (P) The value of m is
(Q) The value of r_1 is
(R) The value of $r_1 + r_2$ is
(S) The value of n is
(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$
(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
(C) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$
(D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

12. Given $f : A \rightarrow B$.



In the List-I there are some conditions and in the List-II there are number of functions satisfying the conditions in the List-I.

Match List-I with List-II and select the correct answer using the code given below the list.

List-I

- (P) $f(1) > f(2) > f(3) > f(4)$
 (Q) $f(i) \neq f(j)$ for $i \neq j$ (where $i, j \in A$)
 (R) $f(1) \geq f(2) \geq f(3) \geq f(4)$
 (S) $f(1) < f(2) < f(3) > f(4)$ and
 $f(i) \neq f(j)$ for $i \neq j$ (where $i, j \in A$)

List-II

- (1) ${}^{10}C_4 \cdot {}^3C_1$
 (2) ${}^{10}C_4 \cdot 24$
 (3) ${}^{10}C_4$
 (4) ${}^{13}C_4$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Two lines are drawn at right angle one being a tangent to $y^2 = 12x$ and the other to $x^2 = 16y$. If the locus of their point of intersection is $(x^2 + y^2)(\ell x + my) + (nx - 3y)^2 = 0$ then $\ell + n - m$ is equal to
- Let AP (a, d) denote the set of all the terms of an arithmetic progression containing 2020 terms with first term a and common difference d. If AP (1, 3) and AP (9, 7) have λ number of uncommon terms, then the value of $\frac{\lambda}{100}$ is

Space for Rough Work

3. If P lies in the first quadrant on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $a > b$), and tangent & normal drawn at P meets major axis at the points T & N respectively, then the value of $\frac{4(|F_2N| + |F_1N|)(|F_2T| - |F_1T|)}{5(|F_2N| - |F_1N|)(|F_2T| + |F_1T|)}$ is equal to
(where F_1 & F_2 are the foci $(ae, 0)$ & $(-ae, 0)$ respectively)
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a polynomial function with real coefficients such that $(x^2 + x + 1) f(x - 1) = (x^2 - x + 1) f(x) \forall x \in \mathbb{R}$ and $f(0) = 1$, then the value of $\frac{f(1)}{10}$ is
5. Let $\text{diag. } (a, b, c)$ represents a 3×3 diagonal matrix with diagonal elements a, b, c (in order). Now, consider matrices $A = \text{diag. } (3, 2, 5)$, $B = \text{diag. } (11, 19, 13)$ and $C = \text{diag. } (61, 53, 31)$. The number of factors of trace of the matrix ' ABC ' is (trace = sum of diagonal elements)

Space for Rough Work

6. If a, b, c are positive real numbers and the minimum value of $\frac{(a+3c)}{(a+c+2b)} + \frac{4b}{(a+b+2c)} - \frac{8c}{(a+b+3c)}$ is m , then the value of $|m|$ is (use $\sqrt{2} = 1.4$)
7. In a tetrahedron $OABC$, the edges are of lengths, $|OA| = |BC| = a, |OB| = |AC| = b, |OC| = |AB| = c$. Let G_1 and G_2 be the centroids of the triangle ABC and AOC such that $OG_1 \perp BG_2$, then the value of $\frac{a^2 + c^2}{b^2}$ is
8. A game of luck is designed in which you have to shoot a target. The organizer puts two bullet in an empty six round revolver and spins. You point it towards balloon and shoot. It was an empty shot (i.e. bullet was not fired). If the bullets are placed randomly then the probability that you will shoot the balloon in next shot is $\frac{P}{Q}$ (Where P and Q are relatively coprimes) {It is assumed that fired bullet always hits the target}, then the value of $(P^2 + Q^2)$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	32
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator